

SMART MOBILITY PLATFORM

Intelligent Road Traffic Monitoring System

IRTMS

A full-stack traffic intelligence platform for monitoring, prediction, route analysis, alerts and analytics.

REAL-TIME MONITORING • AI-ASSISTED PREDICTION • INTELLIGENT ROUTE ANALYSIS

[OPEN LIVE DEMO](#)

[VIEW SOURCE ON GITHUB](#)

PROJECT DOCUMENTATION

Professional technical and product overview • Current deployed implementation

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Quick access

[Live Demo](#)

[GitHub Repository](#)

[Live Dashboard](#)

The PDF also contains clickable external links on the cover and final project-links page.

1. Project Overview

The Intelligent Road Traffic Monitoring System (IRTMS) is a web-based smart mobility platform designed to centralize traffic observation, congestion analysis, prediction, route planning, alerts, and operational analytics.

The application combines a React/Vite frontend with a FastAPI backend, database-backed traffic observations, map visualization, authentication, role management, external traffic-provider integration, and analytical services.

The project is designed as an operational dashboard rather than a static visualization. A user selects a monitoring location, reviews current traffic conditions, investigates trends, evaluates routes, generates alerts, and uses analytical views to understand road performance.

Key capabilities

- Location-based traffic monitoring for Indian states and cities/towns.
- Interactive geographic traffic visualization.
- Congestion classification and road-level summaries.
- Traffic prediction based on supplied road conditions and prediction horizon.
- Traffic-aware route planning with map visualization and route alternatives.
- Operational alert creation and management.
- Historical traffic analytics including trends, heatmaps, road performance and utilization.
- JWT authentication and role-based access control.
- Responsive desktop and mobile application experience.

Live deployment

<https://intelligent-road-traffic-management-u87d.onrender.com>

2. Problem Statement & Objectives

Urban traffic conditions change continuously across roads, corridors and junctions. A useful traffic-management platform therefore needs to bring together geographic monitoring, traffic measurements, prediction, route analysis and operational communication in one interface.

IRTMS addresses this by providing a modular interface in which traffic observations become the common data layer for monitoring and decision-support workflows.

Objectives

- Create a centralized interface for traffic monitoring.
- Represent traffic conditions geographically instead of relying only on tables.
- Convert traffic measurements into understandable congestion levels.
- Support forward-looking traffic prediction.
- Provide route alternatives using geographic routing services.
- Surface operational incidents through alerts.
- Provide historical and comparative analytics.
- Protect operational functions through authentication and roles.

3. System Architecture

IRTMS follows a layered client-server architecture. The frontend is responsible for interaction and visualization, while the FastAPI backend handles business logic, authentication, data normalization, persistence, and external-service communication.

USER INTERFACE	APPLICATION API	DATA / SERVICES
React + Vite React Router Reusable components Leaflet maps Recharts	FastAPI routes Service modules Validation Authentication Business logic	SQLAlchemy / database TomTom traffic APIs Routing / location services Environment configuration

Primary data flow

User → React page → Axios request → FastAPI route → service layer → database / external provider → normalized response → visualization.

The traffic provider credential remains a backend concern; the frontend communicates with the application's API rather than exposing the provider key directly.

Module-level architecture

- Authentication and user-management modules control identity and access.
- Traffic monitoring provides live observation and road-utilization data.
- Route analysis resolves locations and provides traffic-aware routes.
- Traffic prediction processes submitted traffic conditions and returns an expected condition.
- Analytics aggregates observations into trends, heatmaps, road performance and utilization.
- Alerts provide operational notification workflows.

4. Technology Stack

Layer	Technologies
Frontend	React, Vite, JavaScript/JSX, React Router, Axios
Visualization	Leaflet, OpenStreetMap map tiles, Recharts
Backend	Python, FastAPI, Pydantic, SQLAlchemy
Security	JWT authentication, password hashing, role-based access
Data / ML	NumPy, scikit-learn, traffic observation records
External services	TomTom Traffic APIs, geocoding / routing services
Deployment	Docker, GitHub, Render

Why this stack

React provides a component-driven operational interface, while FastAPI gives the system a lightweight asynchronous API layer. Leaflet is used for geographic interaction, Recharts for analytical visualizations, SQLAlchemy for persistence, and Docker/Render for reproducible deployment.

5. Core Functional Modules

Dashboard — Executive traffic overview with monitoring scope, live traffic map, vehicle observations, average speed, congestion indicators, alerts and system status.

Live Monitoring — Focused view of monitored locations, traffic-flow categories, average speed, congestion levels, vehicle counts, monitoring status and mapped points.

Traffic Prediction — Accepts road name, prediction horizon, time/day context, vehicle count, current speed and free-flow speed, then presents predicted congestion and confidence.

Route Planner — Accepts origin and destination, evaluates route alternatives, presents distance/travel-time/speed metrics and draws the selected route on a map.

Alerts — Creates operational traffic alerts using area, road, date/time, congestion level, severity, current speed, free-flow speed and description.

Analytics — Provides monitored-road counts, average speed, historical vehicle observations, traffic trends, congestion distribution, heatmap points, road performance and utilization.

User Management — Allows authorized users to search accounts, filter roles/status, control roles and activate/deactivate accounts.

6. Application Screens & Outputs

The following screenshots are taken from the implemented application and document the visible product output and workflows.



Figure 1 — Secure authentication / login interface

Authentication provides a controlled entry point to the traffic-management dashboard. The UI communicates the platform identity and provides a dedicated sign-in workflow.

— SMART MOBILITY PLATFORM —

Intelligent Road Traffic Monitoring System

Intelligent Traffic Management

Real-time traffic monitoring
AI-powered congestion prediction
Intelligent route analysis

ACCOUNT REGISTRATION

Create your account

Register for access to the Intelligent Road Traffic Monitoring System.

FULL NAME

Enter your full name

EMAIL ADDRESS

Enter your email address

PASSWORD

Create a secure password

CHOOSE YOUR ROLE

Commoner

Commoner accounts can be registered directly.
Commissioner registration requires authorization.

Create account

Already have an account? [Sign in](#)

Account access is controlled by the IRTMS authentication and role management system.

Figure 2 — Account registration and role selection

Registration supports account creation and role selection. The interface distinguishes common-user registration from commissioner authorization requirements.

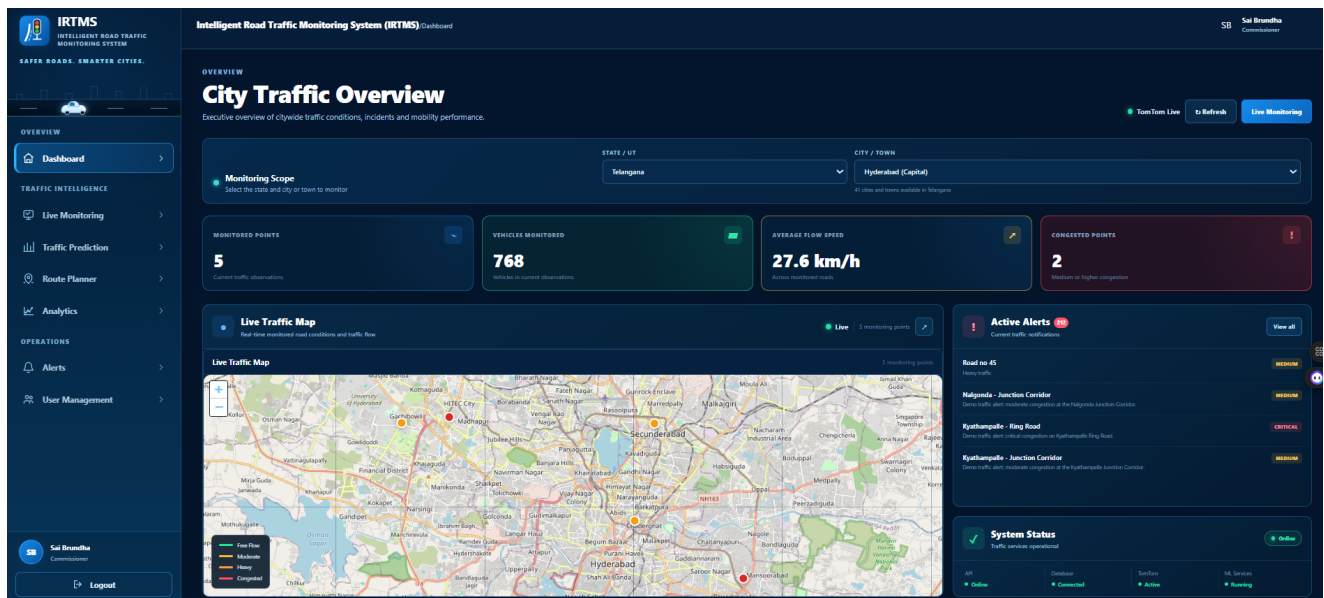


Figure 3 — Executive dashboard with live traffic map, alerts and system status

The dashboard combines the main operational signals in a single view: monitored points, vehicles, average flow speed, congested points, geographic traffic visualization, active alerts and service status.

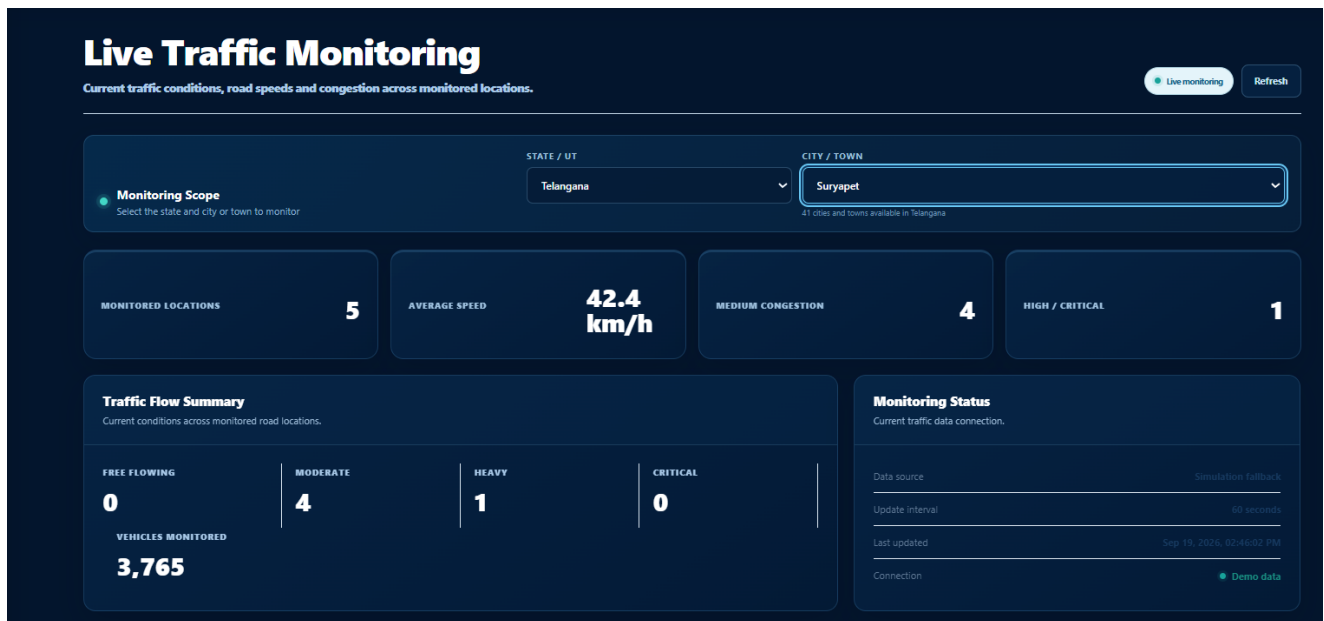


Figure 4 — Live Monitoring: traffic-flow summary and monitoring status

Live Monitoring summarizes the current observation set by traffic-flow condition, average speed, congestion severity and vehicle volume. The monitoring-status panel communicates the current data source and connection state.

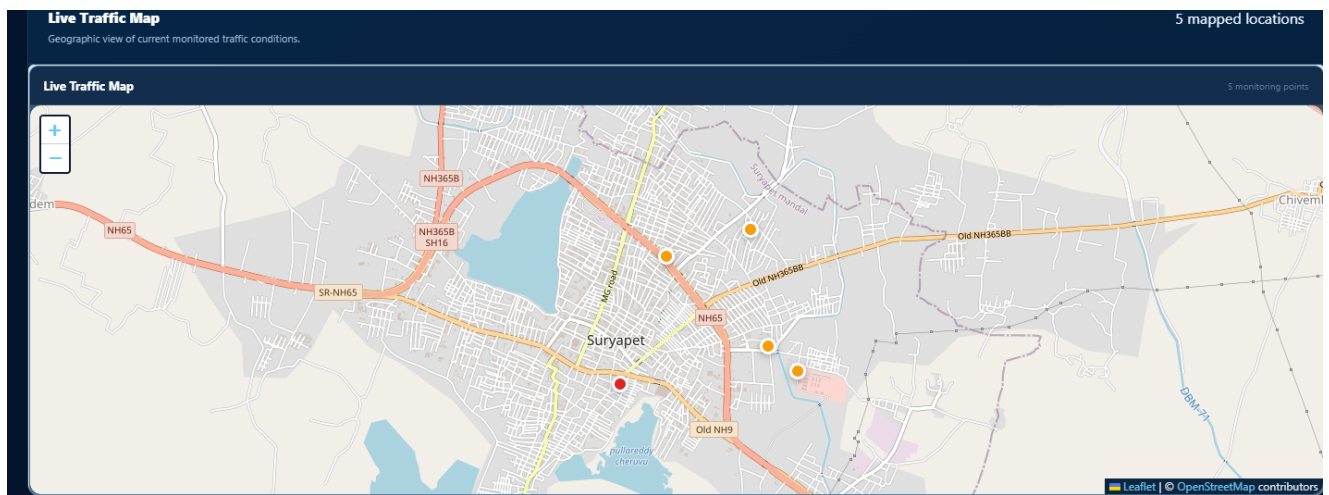


Figure 5 — Live Traffic Map: geographic view of monitored locations

The map visualizes the selected monitoring scope and its five monitored locations. Map labels and road geometry provide geographic context for the traffic observations.

Traffic Prediction

Predict congestion from historical traffic data and current road conditions.

Prediction service

Success. Traffic prediction generated successfully.

Prediction Inputs

Enter the current traffic conditions.

ROAD NAME

Suryapet

PREDICTION HORIZON

30 minutes

HOUR OF DAY

14:00

DAY OF WEEK

Wednesday

VEHICLE COUNT

800

CURRENT SPEED (KM/H)

32

FREE-FLOW SPEED (KM/H)

60

Clear

Predict Traffic

Prediction Result

Expected traffic condition from the submitted conditions.

PREDICTED CONGESTION

Medium

Estimated Speed

25.0 km/h

Confidence

85.0%

Prediction Horizon

30 min

Road

Suryapet

Traffic Assessment

Moderate congestion is expected for the selected horizon.

Input Conditions

Vehicle count800
Current speed32.0 km/h
Free-flow speed60.0 km/h
Time14:00 • Wednesday

Figure 6 — Traffic Prediction: input conditions and prediction result

The prediction interface accepts operational traffic conditions and returns an expected congestion category, estimated speed, confidence, prediction horizon and an explanatory assessment.

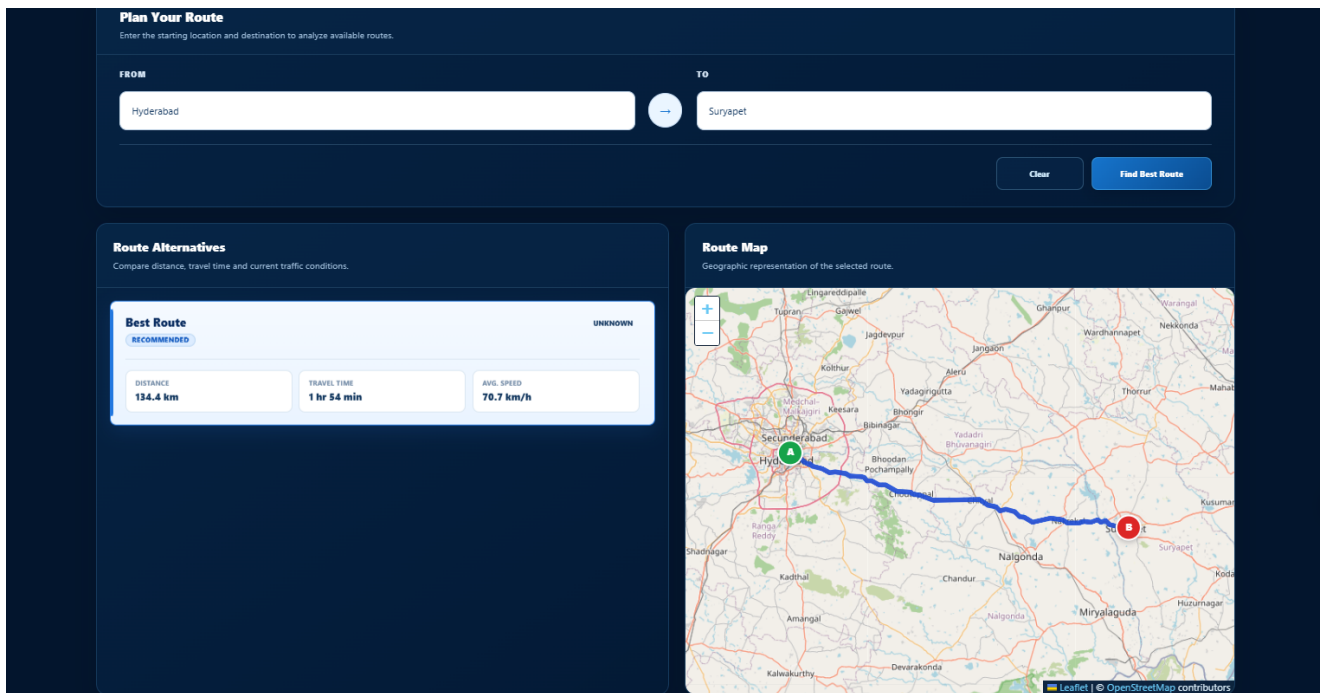


Figure 7 — Route Planner: route alternative and geographic route visualization

The Route Planner combines route metrics with a geographic route map. The selected route displays distance, travel time and average speed, while the map shows the origin-to-destination path.

Traffic Alerts

Review active traffic incidents and operational notifications.

RefreshCreate Alert Form

Create Traffic Alert

Publish a verified traffic condition to the shared alert system.

AREA

Sunnyvale

ROAD

Highway

ALERT DATE & TIME

09/19/2026 02:40 PM

CONGESTION LEVEL

Medium

SEVERITY

Medium

CURRENT SPEED (KM/H)

23.2

FREE-FLOW SPEED (KM/H)

80.56

Derived congestion71.2%Calculated from current speed versus free-flow speed.

DESCRIPTION

Heavy traffic and congestion on highway, try to avoid

Figure 8 — Traffic Alerts: operational alert creation workflow

The alert workflow captures the location, road, timing, congestion level, severity, speed conditions and description required to publish a traffic notification.

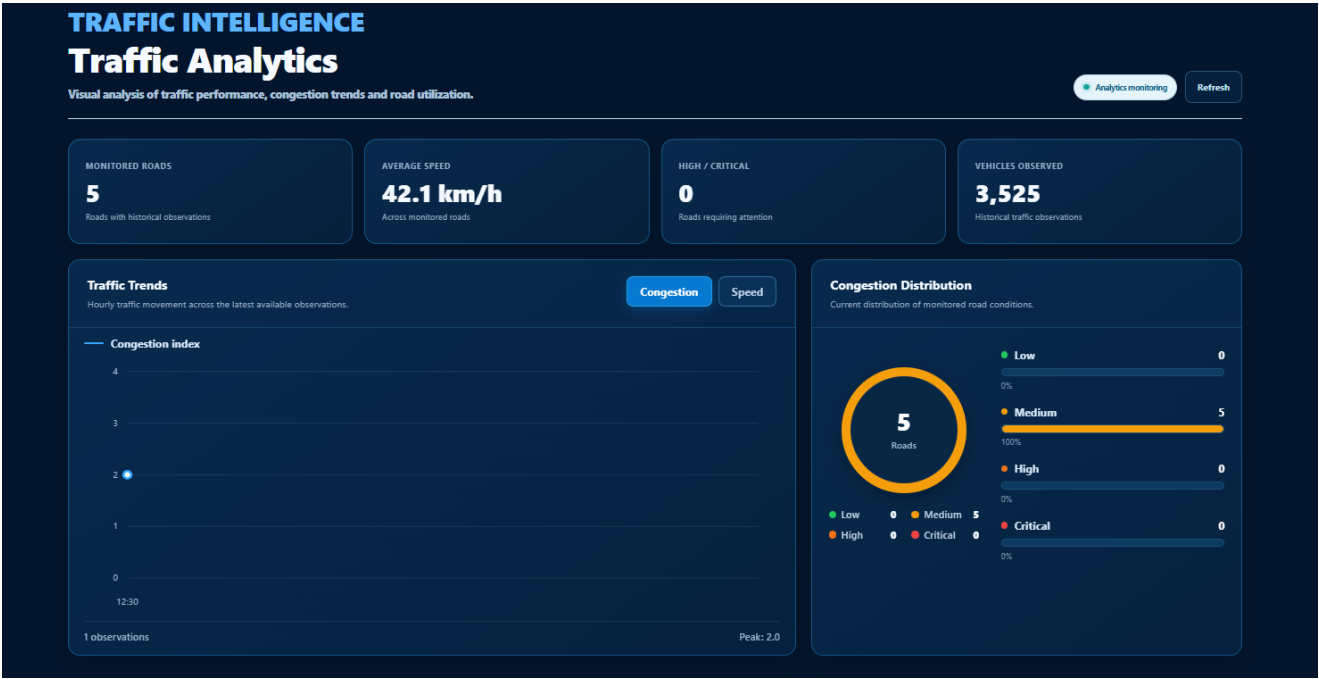


Figure 9 — Traffic Analytics: KPI cards, trends and congestion distribution

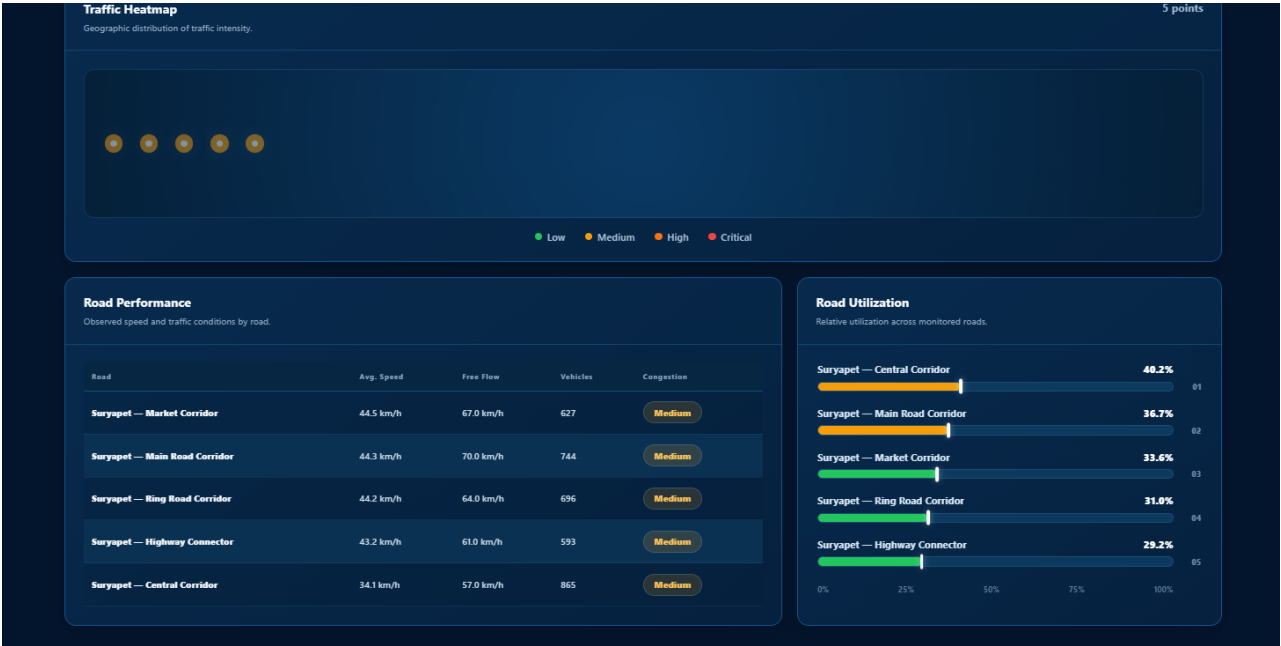


Figure 10 — Traffic Analytics: heatmap, road performance and road utilization

Analytics converts traffic observations into operationally useful summaries. The page combines KPI cards, trend visualization, congestion distribution, geographic heatmap points, road-level performance and utilization.

Registered Users						
Change roles or activate and deactivate user accounts.						
Search by name or email...			All Roles		All Statuses	
User	Email	Role	Registered	Status	Role Control	Account
SM Sai Brundha Maligireddy	maligireddysaibrundha@gmail.com	Commissioner	—	Active	Commissioner	Deactivate
SB Sai Brundha Current account	saibrundha@gmail.com	Commissioner	—	Active	Commissioner	Deactivate
PR Padma Reddy	padma@gmail.com	Commoner	—	Active	Commoner	Deactivate
AM Ananya	ananya@gmail.com	Commoner	—	Active	Commoner	Deactivate
PO Pojitha	24eg105d51@anurag.edu.in	Commoner	—	Active	Commoner	Deactivate
SB Sai Brundha	saibrundha2006@gmail.com	Commissioner	—	Active	Commissioner	Deactivate

Figure 11 — User Management: accounts, roles, status and account controls

The User Management interface provides administrative visibility into registered accounts and exposes role and account-status controls to authorized users.

7. Traffic Data & Intelligence Workflow

The traffic pipeline is designed around a monitoring scope: a selected state and city/town are associated with geographic coordinates, which are then used to request traffic observations.

Live traffic workflow

- Select the state and city/town to monitor.
- Resolve or validate geographic coordinates for the selected monitoring point.
- Request traffic information from the configured traffic provider.
- Normalize provider results into the application's traffic-observation format.
- Persist observations where configured for historical analytics.
- Expose the observations to Dashboard, Live Monitoring and Analytics.

Fallback behavior

The implementation includes a controlled simulation fallback for cases where live provider data is unavailable. This preserves application continuity for demonstrations and development. Simulated observations must be interpreted as demonstration data rather than real-world measurements.

Analytics data path

Persisted traffic observations are aggregated into trend buckets, congestion distributions, heatmap points, road-performance rows and utilization indicators. This gives the same underlying traffic observations multiple operational views.

8. Backend & API Architecture

The backend is organized into domain modules rather than placing all functionality into a single route file. This separation makes traffic monitoring, prediction, routing, analytics, alerts and user management independently maintainable.

/api/auth	Authentication and account access
/api/auth/users	User-management operations
/api/traffic	Live traffic, observations and utilization
/api/prediction	Traffic prediction
/api/routes	Location search and route analysis
/api/alerts	Traffic alerts
/api/analytics	Heatmap, trends, road performance and summaries
/api/workflow	Workflow-related backend operations

Operational health

The backend exposes health and API documentation endpoints for deployment verification and developer inspection. The application can therefore be checked independently of the frontend during deployment troubleshooting.

9. Security, Roles & Access

- JWT-based authentication is used to protect authenticated application workflows.
- Password hashing is used rather than storing plaintext passwords.
- Protected frontend routes separate public authentication screens from operational pages.
- Role-based controls distinguish common users from commissioner-level administration.
- Sensitive provider credentials are intended to remain in backend environment variables.
- Administrative account controls include role management and account activation/deactivation.

The application therefore treats authentication, authorization and operational data access as separate concerns rather than relying only on frontend visibility.

10. Deployment & DevOps

The project is maintained in GitHub and deployed to Render. Docker support provides a reproducible backend/runtime environment and simplifies local-to-cloud parity.

Typical development lifecycle

Develop → test locally → verify API → build frontend → validate Docker services → commit to GitHub → deploy to Render → smoke-test the live application.

Production configuration

Sensitive configuration is supplied through environment variables. Typical values include the database connection, JWT secret, token configuration, TomTom API key and frontend API URL.

Live resources

[Live Application](#)

[GitHub Repository](#)

11. Testing & Validation

The implemented workflow has been validated through end-to-end UI operation and deployment troubleshooting across the principal modules.

- Authentication screens render and support the login/registration workflow.
- Monitoring scope selection drives geographic traffic visualization.
- Traffic maps render geographic basemap tiles and monitoring points.
- Traffic prediction returns a structured result with congestion, speed and confidence information.
- Route planning renders a selected route and associated route metrics.
- Alert creation captures traffic conditions and severity.
- Analytics displays traffic observations across multiple visualization types.
- User Management exposes account and role controls to authorized users.
- Responsive navigation has been implemented for smaller screens.

12. Current Limitations

- Live traffic availability depends on the configured external provider and its API access, quotas and response quality.
- When provider data is unavailable, the application can use simulation fallback for continuity; those observations are not live measurements.

- Prediction quality is dependent on the quantity and quality of historical observations available to the prediction workflow.
- Production analytics benefit from a persistent database with a growing history of observations.
- External routing and geocoding services can introduce latency or availability dependencies.

13. Future Enhancements

- Build a larger historical traffic dataset and automated model-retraining pipeline.
- Add incident, road-closure and accident intelligence.
- Add traffic-signal optimization and recommended intervention workflows.
- Introduce richer audit logs and operational observability.
- Expand automated frontend and backend test coverage.
- Add richer mobile-specific analytics and low-connectivity behavior.
- Introduce configurable traffic corridors and user-defined monitoring zones.
- Add exportable reports for traffic authorities and operational teams.

14. Project Links & Conclusion

Project links

Live Demo — <https://intelligent-road-traffic-management-u87d.onrender.com>

Dashboard — <https://intelligent-road-traffic-management-u87d.onrender.com/dashboard>

GitHub — [bru-reddy/Intelligent_Road_Traffic_Management_System](https://github.com/bru-reddy/Intelligent_Road_Traffic_Management_System)

Conclusion

IRTMS demonstrates a complete smart-mobility application workflow: authenticated access, location-aware traffic monitoring, geographic visualization, prediction, route analysis, alerts, analytics and role-based administration. Its modular architecture provides a foundation for expanding from a project prototype into a more data-rich traffic intelligence platform.

This document combines the technical architecture with screenshots of the implemented interface so that the project can be evaluated both as a software system and as a working product.

Thank you for reviewing the Intelligent Road Traffic Monitoring System.